

Site Selection for Startup Investment Firm in Michigan

Introduction: This report synthesizes data from a 2015 Financial Industry Regulatory Authority study on taxable account investment behavior, along with datasets from the U.S. Census Bureau, the U.S. Geological Survey, open data provided by the state of Michigan, and construction site selection guidance from Building Advisor related to slope and its impact on development costs. Integrating financial, demographic, geographic, and construction considerations, the analysis identifies Oakland and Livingston counties as optimal locations for establishing an investment firm.

Investment Behavior & Demographics: Higher household incomes (above \$50,000) and attainment of a college education—particularly when achieved after marriage—were identified as key factors influencing the likelihood of owning taxable investment accounts (Figure 1) (Mottola, 2015).

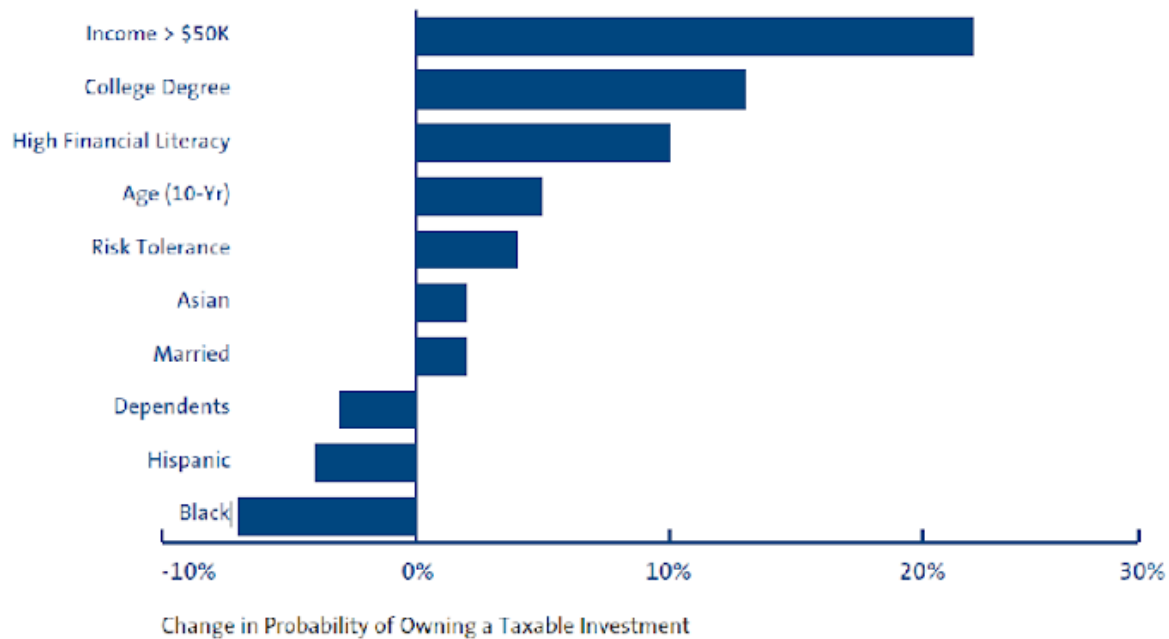
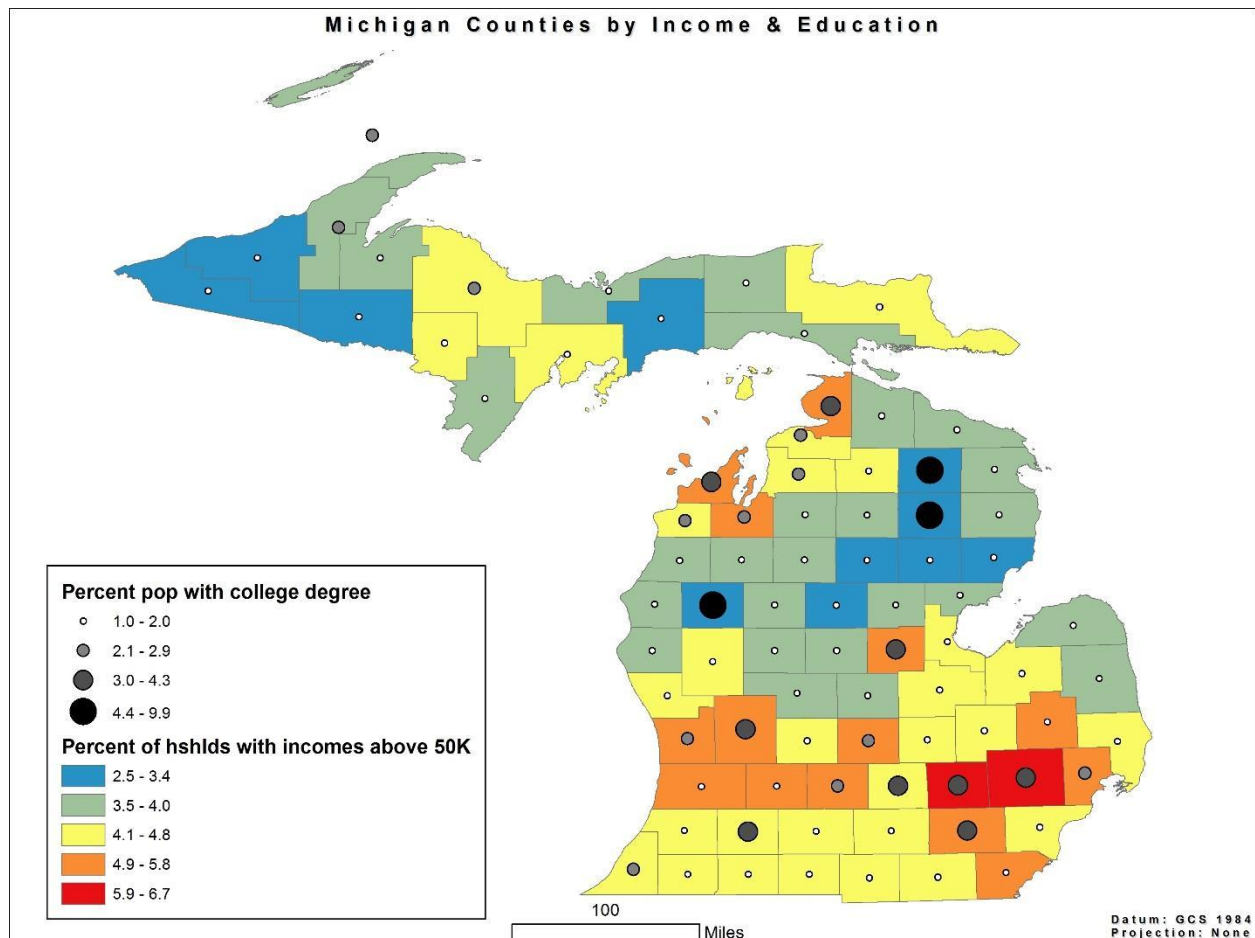


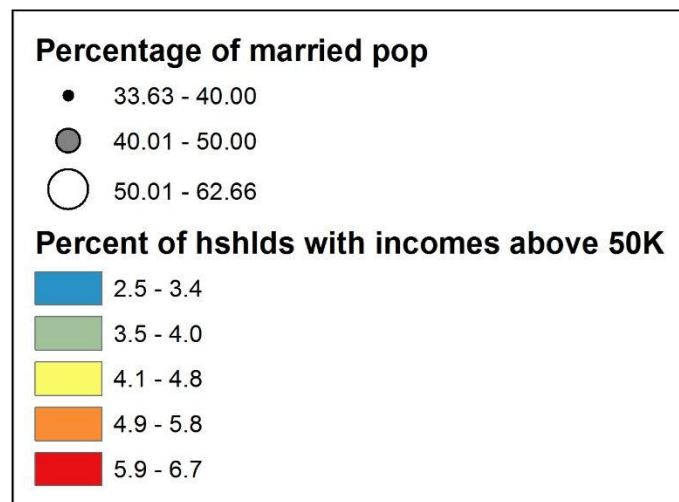
Figure 1: 2015 FINRA study table on taxable account ownership probability.

Based on findings from the Financial Industry Regulatory Authority study, U.S. Census data were analyzed and joined to four vector datasets representing income, education, marriage rates, and population. The highest percentages of households earning \$50,000 or more were concentrated in Oakland and Livingston counties (Map 1). These counties also ranked in the

third-highest quartile for educational attainment. Income, education, and marriage rates were analyzed. Oakland County emerged as the wealthiest county in Michigan. While marriage rates in Oakland and Livingston counties were moderate, they were not among the highest observed (Map 2).



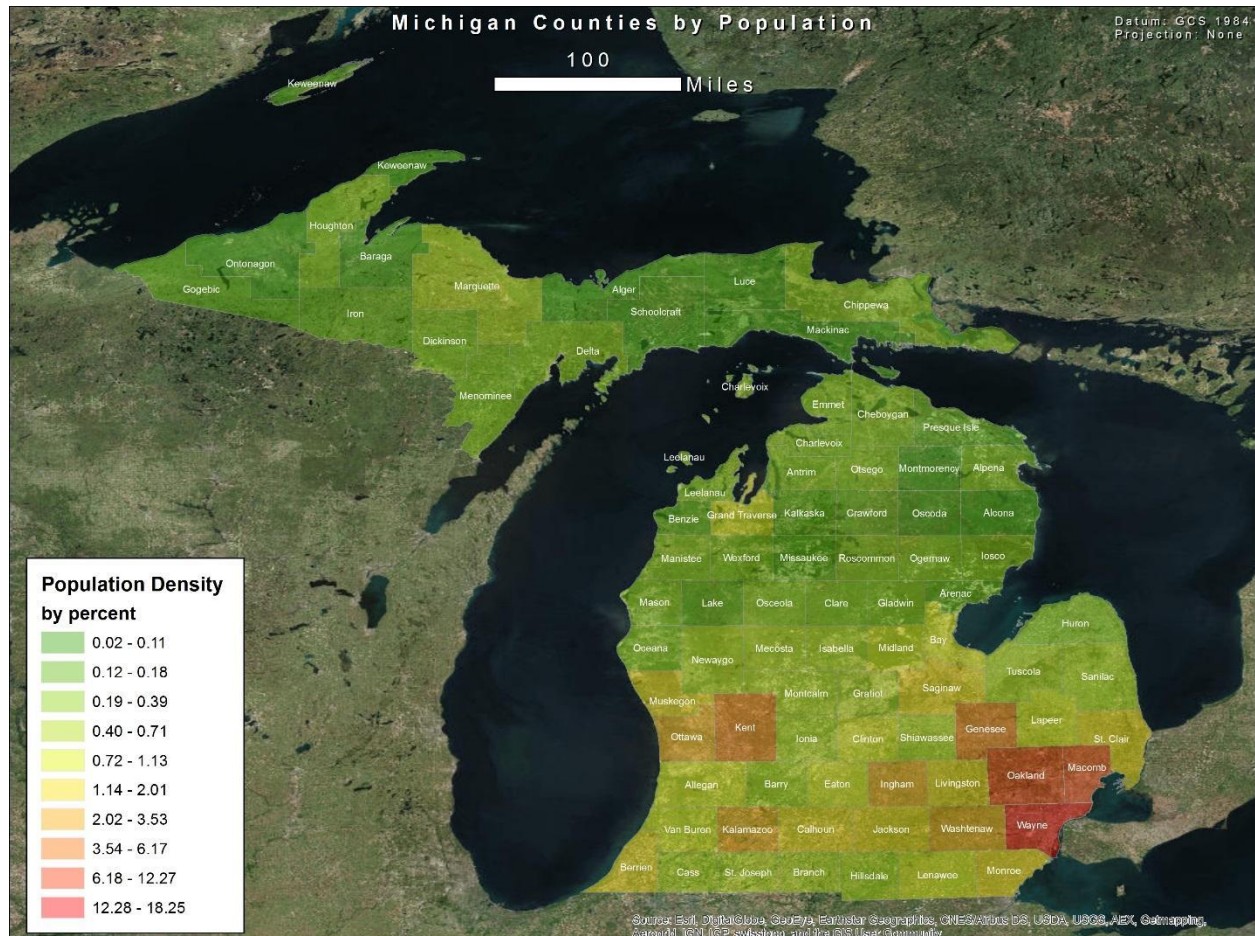
Michigan Counties by Married & Income



100
Miles

Datum: GCS 1984
Projection: None

Population: Wayne County contains the largest share of Michigan's population, followed by Oakland County (Map 3). The substantial population difference between Oakland and Livingston counties may help explain why Oakland has a greater number of households with higher average incomes.

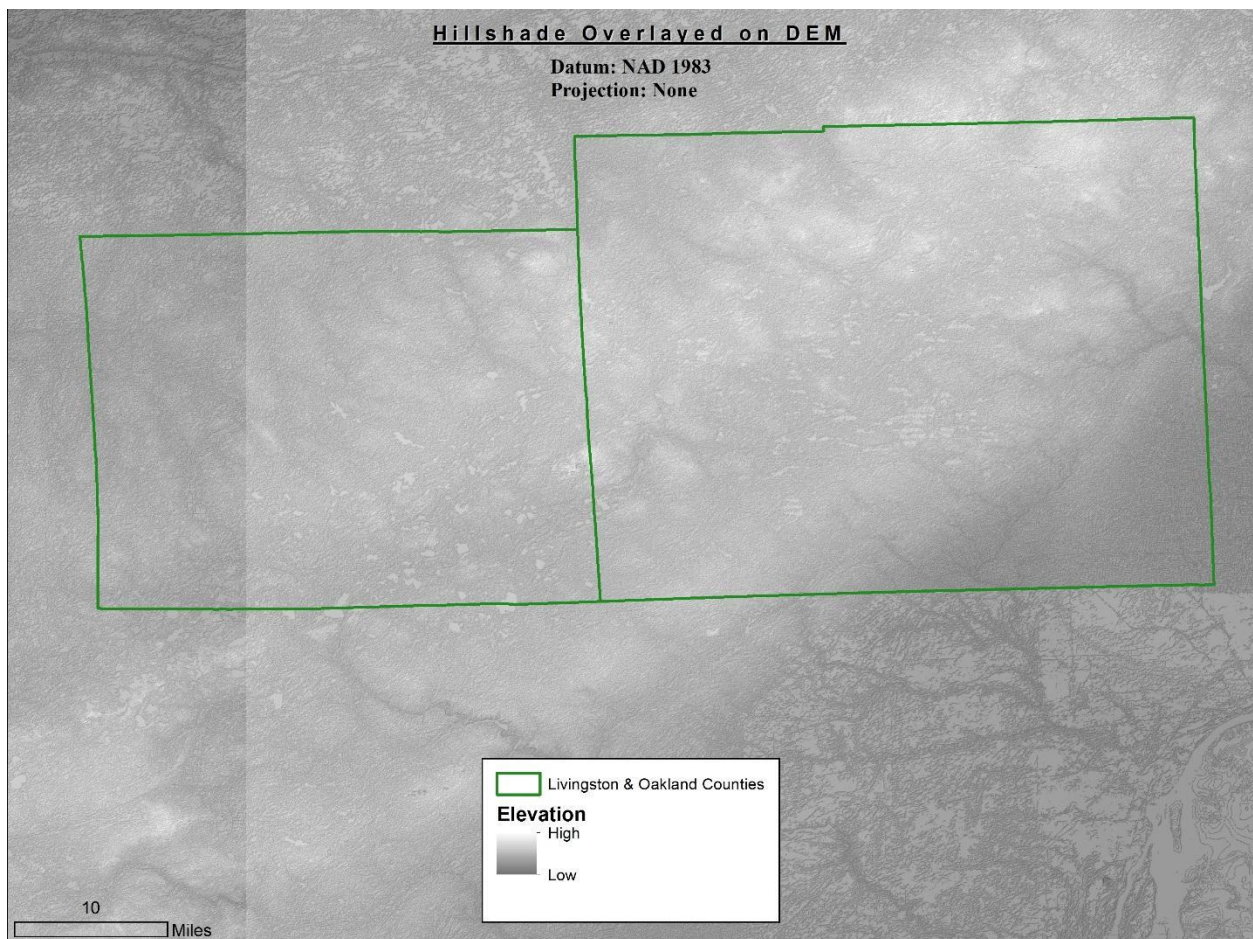


Site Selection: Between Oakland and Livingston counties, Oakland County was selected as the primary location for establishing an investment firm due to its higher percentage of households with elevated average incomes, moderate levels of college degree attainment, and larger population base. Livingston County was identified as a secondary, alternative location for potential firm establishment.

Site Selection Methodology: Slope, land cover, and road network data were integrated to identify suitable construction sites within the selected counties. Two 1×1 -degree digital elevation models (10-meter resolution) from the U.S. Geological Survey were acquired (Map 4), reprojected, and processed to identify areas with slopes less than 10% (Map 5), thereby

minimizing construction costs. As noted by BuildingAdvisor, “Flat sites or gentle grades of less than about 10% (a one-foot rise over 10 feet) are the easiest to build on. Between 10% and 20%, additional earthmoving and grading are typically required, increasing overall construction costs” (BuildingAdvisor.com, 2016).

Land cover data (2011) were then analyzed to identify areas classified as barren land and developed open space, which are more suitable for construction (Map 6). Finally, road network data from the state of Michigan were evaluated. Buffers of 500 meters for primary roads and 25 meters for secondary roads were generated and converted into raster format to assess accessibility. Following the integration of slope, land cover, and road proximity criteria, areas were classified as suitable and unsuitable for development based on the defined parameters (Map 7).



Percent Slope

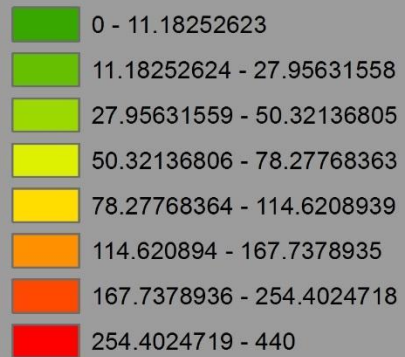
Datum: NAD 1983

Projection:Hotline Oblique Mercator GeoRef

Cty_Focus_P

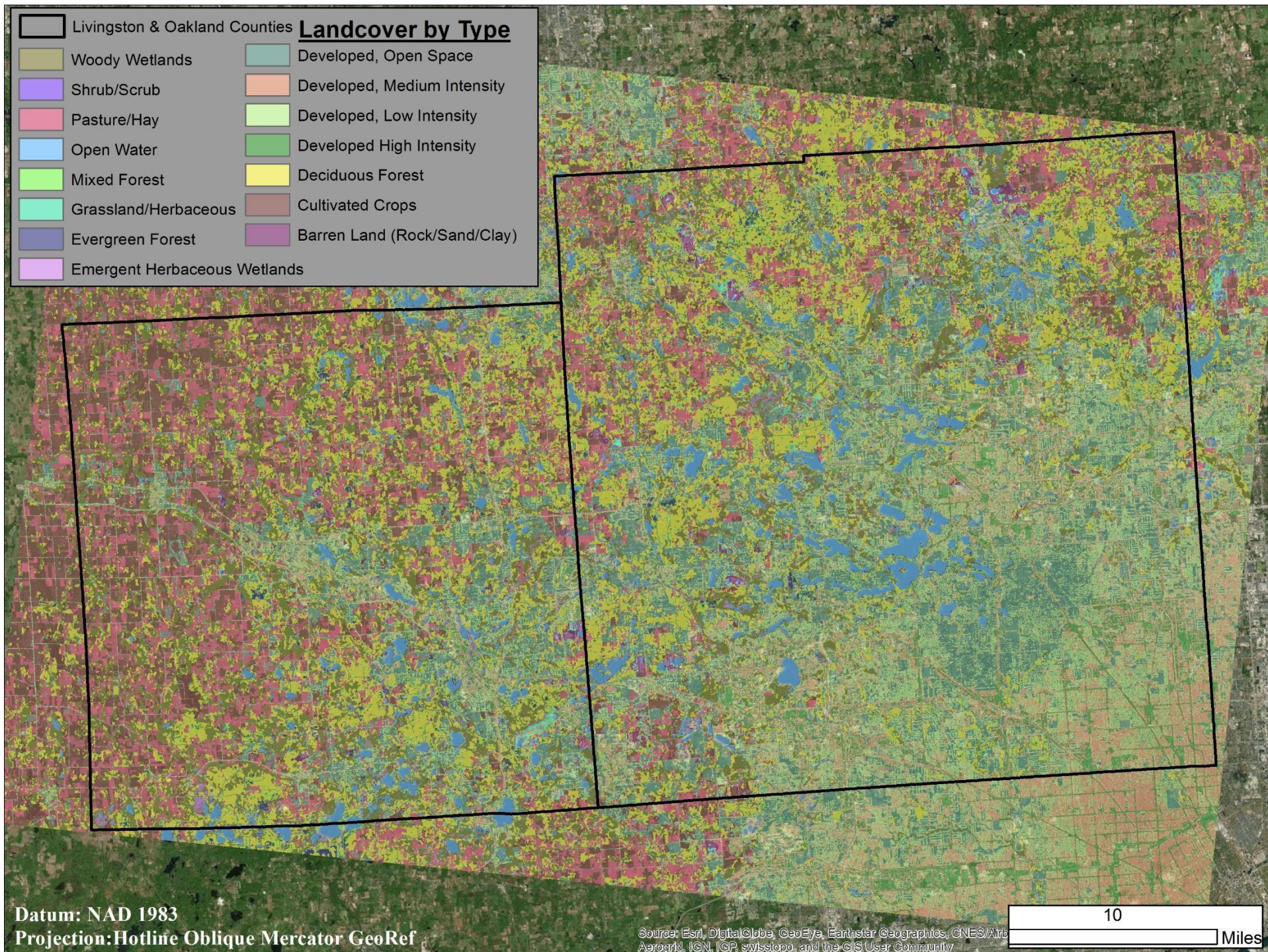
Percent Slope

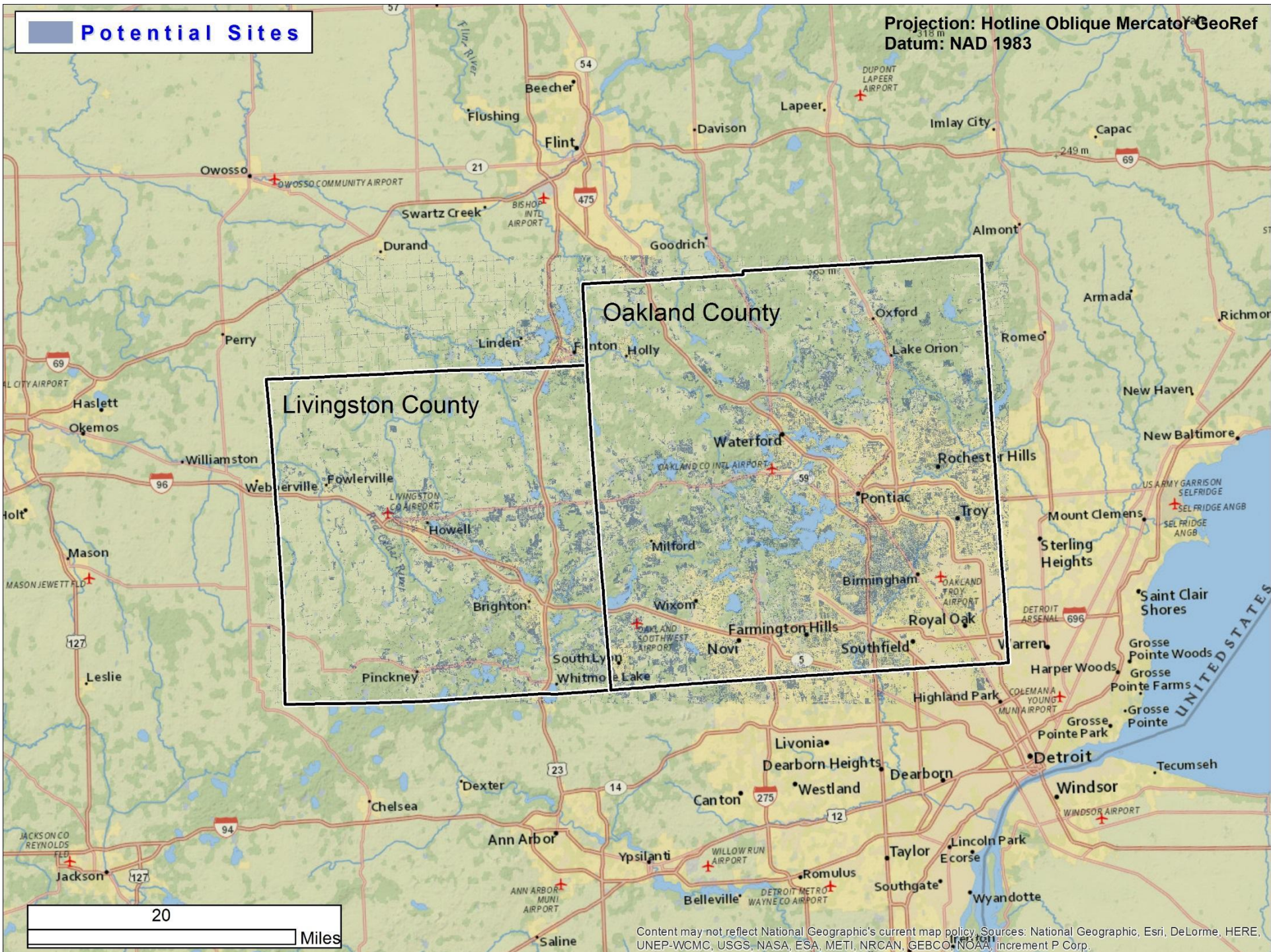
Value in %



10

Miles



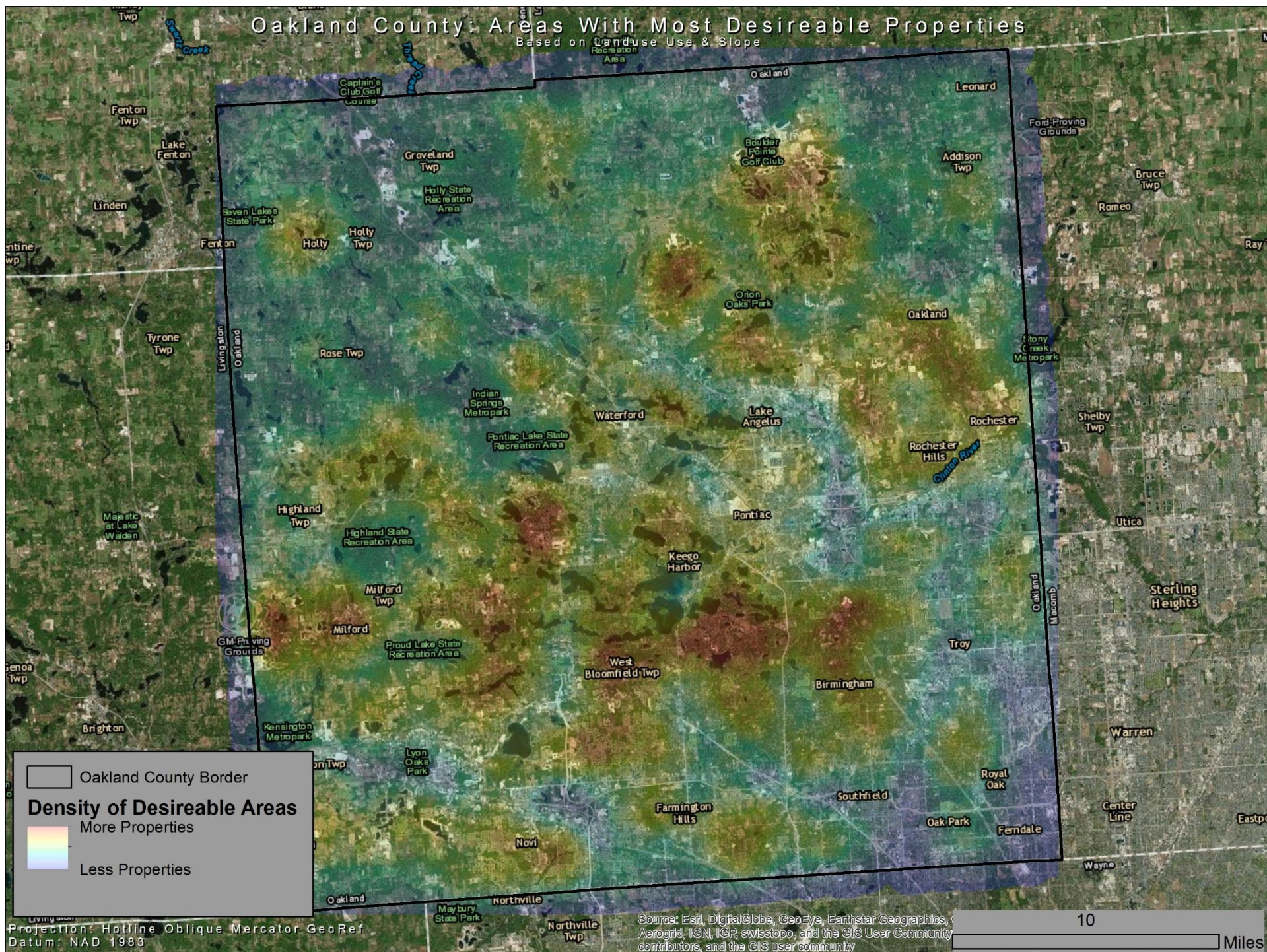


Parcel-level data—which would have significantly enhanced the site selection analysis—were not available for Oakland and Livingston counties. To address this limitation, three processing steps were applied to the identified suitable area polygons. First, polygons with areas between 3,000 and 20,000 square feet were selected to approximate viable parcel sizes. Second, point features were generated from these polygons to represent candidate locations. Finally, a point density surface (per square mile) was created to further refine the identification of suitable areas. As an additional benefit, the use of point density analysis reduced visual clutter and improved the interpretability of the Oakland and Livingston maps.

Site Selection Analysis: In addition to favorable demographics for taxable investment accounts, Oakland County (Map 8) offered nine potential candidate areas for site selection, compared to five in Livingston County (Map 9). In total, approximately 14 candidate areas were identified across both counties. Oakland County’s greater number of viable sites reflects its higher level of development relative to Livingston County, which remains more rural. Consequently, Oakland County is the preferred location for establishing an investment firm. However, Livingston County represents a suitable alternative, primarily due to its strong concentration of households with incomes exceeding \$50,000 - the second highest among the counties analyzed.

Oakland County: Areas With Most Desirable Properties

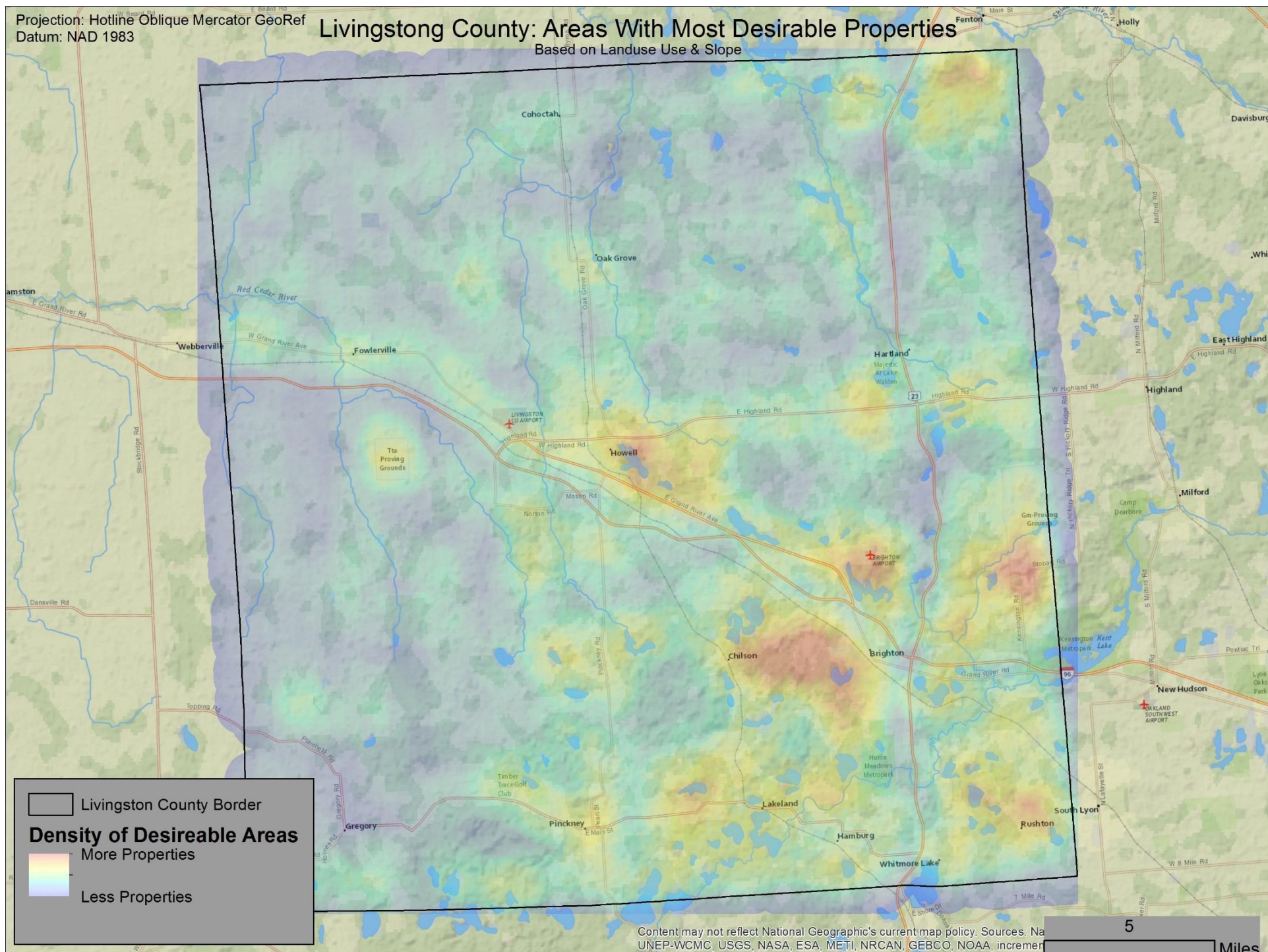
Based on Land Use & Slope



Projection: Hotline Oblique Mercator GeoRef
Datum: NAD 1983

Livingstong County: Areas With Most Desirable Properties

Based on Landuse Use & Slope



Works Cited

ESRI Imagery Basemap, World Imagery, Source: Esri, DigitalGlobe, GeoEye, Earthstar Geographics, CNES/Airbus DS, USDA, USGS, AEX, Getmapping, Aerogrid, IGN, IGP, swisstopo, and the GIS User Community

ESRI National Geographic Basemap, National Geographic, Sources: National Geographic, Esri, DeLorme, HERE, UNEP-WCMC, USGS, NASA, ESA, METI, NRCAN, GEBCO, NOAA, increment P Corp.

“Evaluating Slope, Soils, Drainage & Radon”, Building Advisor website,
<http://buildingadvisor.com/buying-land/site-characteristics/slope-soils-water-vegetation/#jump1>

Mottola, Gary, Insights: Financial Capability “A Snapshot of Investor Households in America,” FINRA Investor Education Foundation. September 2015

National Elevation Dataset, U.S. Geological Survey, The National Map,
<http://viewer.nationalmap.gov/basic/?basemap=b1&category=ned,nedsrc&title=3DEP%20View>

National Land Cover Database 2011, U.S. Geological Survey, Earth Resources Observation and Science (EROS) Center MRLC Project, Sioux Falls, South Dakota, <http://www.mrlc.gov>

State of Michigan; GIS Open Data, Boundary & Roads Data, URL:
<http://gis.michigan.opendata.arcgis.com/>;

US Census Bureau; American Community Survey, 2013 American Community Survey 5-Year Estimates; using American FactFinder; URL: <http://www.census.gov/geo/maps-data/data/tiger-data.html>